



BITS Pilani
Pilani Campus



SURVEYING (CE F213)

(Introduction)

Text Books:



- T1. Duggal S.K.; Surveying; Tata McGraw-Hill, New Delhi, Vol I and II, 4th Edition (2013).
- T2. Moondra, H.S. and Gupta Rajiv, Lab Manual for Civil Engineering, CBS, 2nd edition, (2000).

Components of the course

- Lecture
- Practical
 1. The theory behind the experiments will be explained.
 2. Conduct the experiments.
 3. Submit handwritten reports.
 4. One/Two Lab quizzes will be taken.
- Tutorial
 1. Doubts will be cleared, numerical will be solved. Test will also be there. The date of the tutorial tests will be notified earlier.

Evaluation Scheme

S.No.	Evaluation Component	Duration	Weightage (%)	Date & Time	Nature of Component
1	Mid Sem	90 minutes	25	As per time table	Closed Book
2	Tutorials (Best 4 out of 6)	50 min	10 (=4x 2.5)	contineous	Closed/Open Book
3	Compre. Exam	3 hours	45	As per time table	Closed Book
4	Lab Quiz	-----	10		Closed/Open Book
5	Lab Record(Hand Written)	-----	10		Open Book

- **Total marks for evaluation:** 300 marks
- **Chamber Consultation Hour:** Thursday 4:00PM to 5:00PM

Introduction to the basic concepts of Surveying

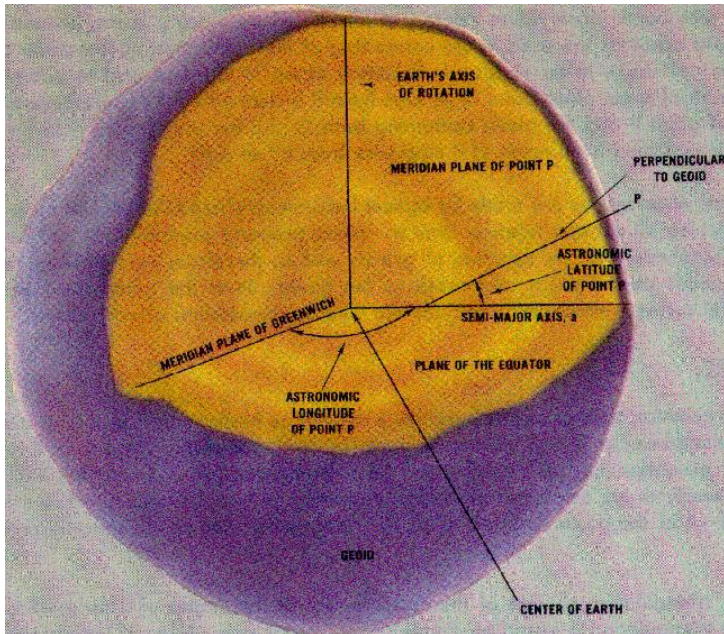


- What is surveying ?
- Surveying is the art of determining the relative positions of the desired points(natural or man-made features) on, above or below the surface of the earth with respect to some other points(benchmark) by measuring distances(horizontal/vertical), angles and directions.
- Objective of Surveying
- It is done to represent the determined information on a sheet as countered map, plan or digital plan(Digital Ground Model: DGM).
- To layout the position of proposed structure for new construction.
- To facilitate other purpose(Like estimation of quantiles etc.)

• Types of Surveying

On the basis of whether the curvature of the earth is taken into account or not during measurement, surveying can be divided into two main categories:

1. Plane Surveying is the type of surveying where the mean surface of the earth is considered as a plane. All angles are considered to be plane angles. For small areas less than 250 km^2 plane surveying can safely be used. For most engineering projects such as canal, railway, highway, building, pipeline, etc. constructions, this type of surveying is used. It is worth noting that the difference between an arc distance of 18.5 km and the subtended chord lying in the earth's surface is 7mm. Also the sum of the angles of a plane triangle and the sum of the angles in a spherical triangle differ by 1 second for a triangle on the earth's surface having an area of 196 km^2 .
2. Geodetic Surveying is that branch of surveying, which takes into account the true shape of the earth (spheroid).

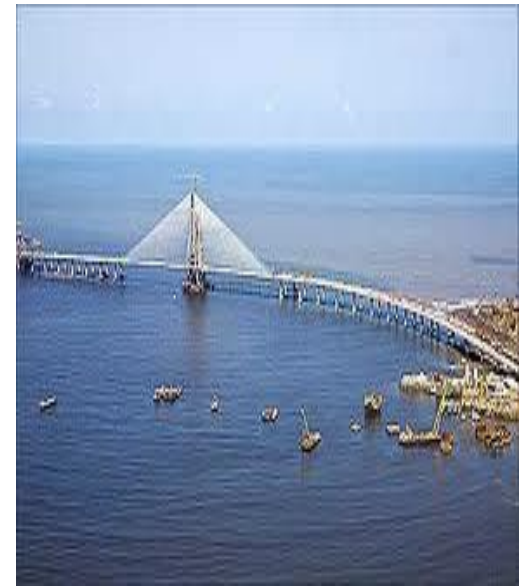


Classification of Survey

- Depending on nature of field(**Place of Survey**)
- Depending on purpose
- Depending on instrument
- Depending on methods used

Depending on nature of field(**Place of Survey**)

1. Land Survey --Topographic survey
 - Cadastral survey(Public land Survey)
 - City survey
2. Hydrographic survey (Marine survey)
3. Underground survey
4. Arial survey
5. Astronomical Survey



Depending on purpose

1. Engineering survey
2. Defence survey
3. Mine survey
4. Geological survey
5. Geographical survey
6. Archeological survey
7. Route survey (*operation sequence: Reconnaissance survey, Preliminary survey, Control survey, Location survey*)

Depending on instrument

1. Chain survey
2. Compass survey
3. Theodolite Survey
4. Levelling
5. Plane Table Survey
6. Triangulation
7. Traversing
8. Tacheometry
9. EDM survey (Electromagnetic Distance Measurement)
10. Total Station Survey
11. Remote sensing, Satellite-based Survey



Depending on methods used

1. Triangulation (Entire area is divided into network of triangles)
2. Traversing (Open traverse, Closed traverse)

Nature of work involved in surveying

The work of a surveyor can be divided in two division

- Field work
- Office work

Field work

1. Adjustment and care of instruments
2. Surveying measurement
3. Recording field notes
4. Field book(keeping all field notes)

Office work

To carryout necessary computations to transfer filed data into a presentable form for plotting.

Principle of surveying

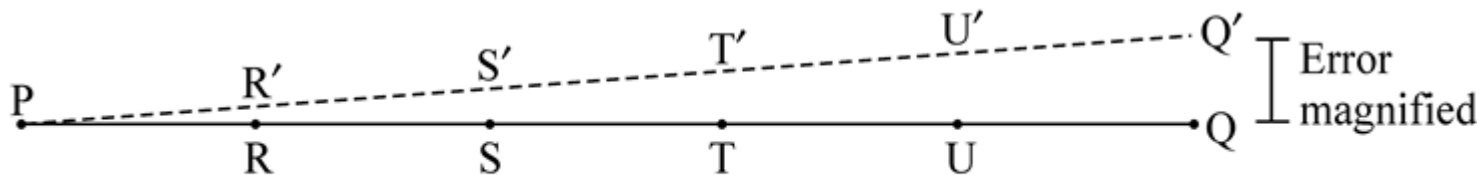
- To work from whole to part
- To locate a point by at least two measurements

To work from whole to part

1. It is the basic principle. This principle should not be violated unless there is no alternative.
2. This is because to localize errors and to prevent their accumulation.



(c) Working from whole to part, error localized



(b) Working from part to whole

To locate a point by at least two measurements

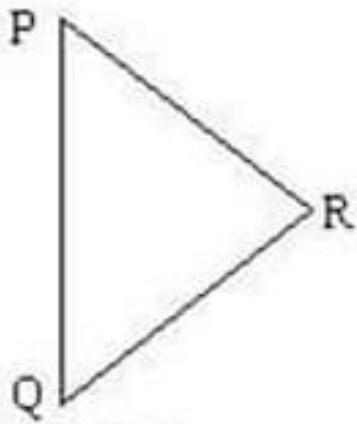


Fig. a

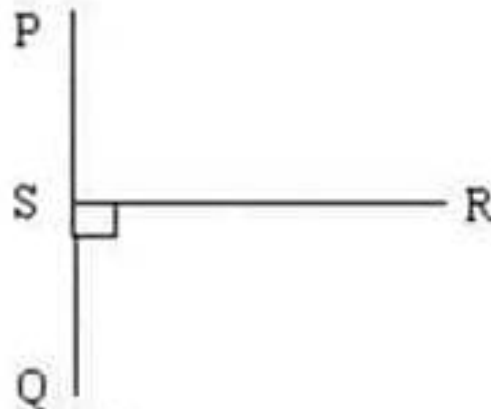


Fig. b

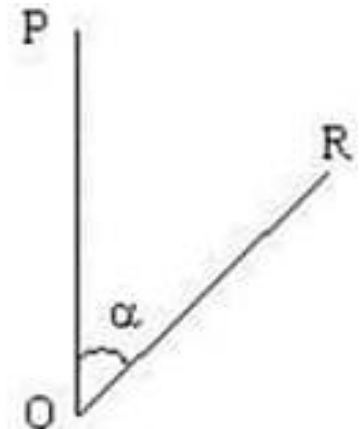


Fig. c

Scale



- It is the fixed proportion between distances of points on map to the corresponding distances of points on the ground.
1. Verbal scale (1cm=1km)
 2. Representative fraction(R.F.)=(1: 100000) for 1cm=1km
 3. Graphical scale---(Scale is drawn in the map itself)- has advantage in shrinkage of map.

Use of shrunk map in original distance calculation



Shrinkage Ratio(SR) or Shrinkage Factor(SF)

$$SF = SR = \frac{\text{Shrunk length}}{\text{Original length}} = \frac{\text{Shrunk scale}}{\text{Original scale}} = \frac{\text{shrunk RF}}{\text{Original RF}}$$

Q1. A surveyor calculated a distance between A and B as 750m using a map which was drawn in a scale of 1cm=50m (RF=1:5000). Later the surveyor found that he had taken wrong scale of 1cm=40m (RF=1:4000) during his calculation. Find the actual length between A and B on the ground.

Q2. An area of field is found to be 80 km² by a planimeter. It is found by using shrunk map, originally drawn with a scale of 1cm=1km. The shrunk map measures 10cm line as 9.75cm. Find the actual area of the field.

Q1. Solution,

Length on the Map is $750/40=18.75\text{cm}$

Length on the ground= $18.75*50=937.5\text{m}$

Q2. Solution,

Shrinkage Factor= $9.75/10=0.975$

So, actual area= $80/(0.975)^2 = 84.155 \text{ m}^2$

